

Module 1: Introduction to Penetration Testing



Topic 1: Introduction to Penetration Testing



What is Penetration Testing?

Penetration testing (also called pen testing) is the practice of testing a computer system, network or Web application to find vulnerabilities that an attacker can exploit.



Penetration Testing Steps

- Information Gathering ->> Passive & Active
- Vulnerability Identification ->> Nmap - Nessus - OpenVas
- Exploitation ->> Brute Forcing - Remote Exploit
- Post Exploitation ->> Persistence - Anti Forensics

Penetration Testing Methodologies

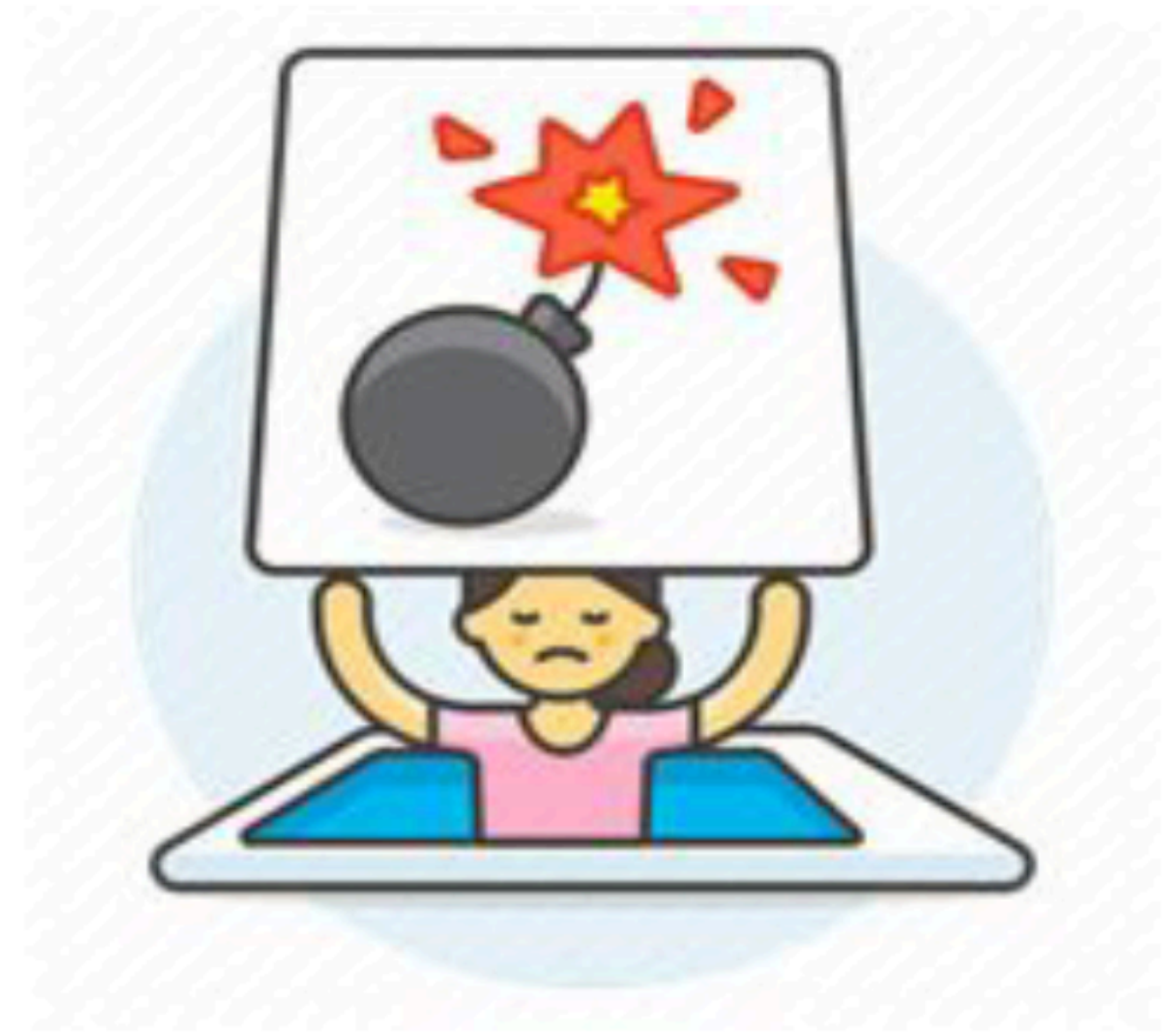
Announced

Unannounced

Black Box Testing (Blind Testing)

Gray Box

White Box Testing (Have Access to Software / Source Code)



What is a Vulnerable System?

A vulnerable system suffer from a misconfigured program, code structure issue or language issue.

Examples:

- SQL Injection -> code structure issue
- Open Port -> Misconfigured Services



Penetration Testing Areas

- **Network/Infrastructure penetration testing**
 - Unpatched OS
 - Misconfiguration of server or network device
- **Web apps penetration testing**
 - Server-side vulnerabilities
 - Client-side vulnerabilities
- **Wireless Network penetration testing**
 - Open/Rogue AP
 - Insecure encryption
- **Social Engineering penetration testing**
 - Dropping USB
 - Unauthorized access
 - Phishing Email with a malicious link

Some areas that can benefit from an understanding of penetration testing are as follows:

- Security operations center (SOC) analysts
- Purple teams
- Network security analysts and engineers
- Application security
- Digital forensics and incident response (DFIR)